

Activity 11 Matrix arithmetic

Aim: Perform matrix calculations on ClassPad.

Enter and store the following matrices:

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$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} -1 & 3 \\ 2 & -2 \\ 5 & 1 \end{bmatrix}, \mathbf{D} = \begin{bmatrix} 1 & 2 & -2 \\ 4 & 2 & -1 \\ 3 & -1 & 2 \end{bmatrix}$$

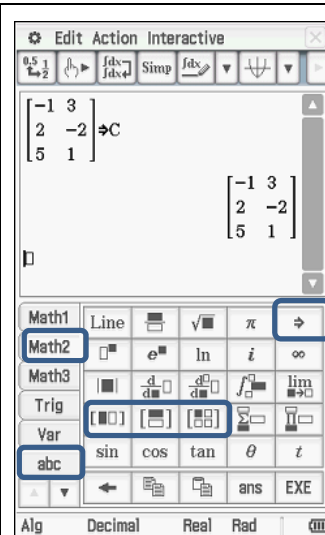
- Tap $\sqrt{x}$ ^{Main}
- Press **Keyboard**, tap **Math2**

Enter matrix C

Use the matrix commands, for example to enter a 3×2 matrix (3 rows and 2 columns):

- Tap $\begin{bmatrix} \square & \square \end{bmatrix}$.
- Tap $\begin{bmatrix} \square \\ \square \end{bmatrix}$.
- Enter the values using the arrow keys to navigate between elements.
- Tap $\Rightarrow$, tap **abc**.
- Tap $\uparrow$, tap C.
- Press **EXE**.
- Tap $\leftarrow$ to go back to the Math menu

Enter and store matrices A, B, D



1. Perform the following calculations (if possible), and record the output. Where a calculation is not possible, explain why.

a) $A + B$

b) $3A$

c) $2B$

d) $2B + 3A$

e) $A + C$

f) $B + 2D$

g) $A \times B$

h) $B \times A$

i) BC

j) $D \times C$

k) $C \times D$

l) A^2

m) $C - B \times A$

Learning notes

Matrices can only be added (or subtracted) when they have the same size, i.e. the same number of rows and the same number of columns.

For multiplication the number of columns in the first matrix must be the same as the number of rows in the second matrix. A reason for defining matrix multiplication in this way is demonstrated in the next activity.